

Nama : Try Asputri Pane

NPP : P067760

Link Repository :

https://github.com/tryaspane/final_project_data_pipeline_bni_corpu.git

1. Transaction Analytics

https://github.com/tryaspane/final_project_data_pipeline_bni_corpu/blob/2790a96ed1a2e8e6fa443562b23f41d69c1fd532/query-training/transaction_analytics.sql

○ Volume & Nilai Transaksi per Channel per Hari

#	full_date	channel_name	volume_transaksi	nilai_transaksi
P	1 2023-01-01	ATM	9	39748012.00
	2 2023-01-01	Call Center / IVR	2	1624863.00
	3 2023-01-01	EDC / Mesin Kasir	3	11208643.00
	4 2023-01-01	Internet Banking	6	14783432.00
	5 2023-01-01	Mobile Banking	14	66977823.00
B	6 2023-01-01	Teller	13	58678242.00
	7 2023-01-02	ATM	3	9479945.00
	8 2023-01-02	Call Center / IVR	4	11058289.00
	9 2023-01-02	EDC / Mesin Kasir	1	1214722.00
	10 2023-01-02	Internet Banking	10	29257706.00
	11 2023-01-02	Mobile Banking	13	53328294.00
	12 2023-01-02	Teller	7	15815003.00
	13 2023-01-03	ATM	10	34281900.00
	14 2023-01-03	Call Center / IVR	4	11843338.00
	15 2023-01-03	EDC / Mesin Kasir	3	8861612.00
	16 2023-01-03	Internet Banking	3	25930075.00

○ Volume & Nilai Transaksi per Channel per Minggu

#	year	month_name	channel_name	volume_transaksi	nilai_transaksi
P	1 2023	Januari	Mobile Banking	486	1753588234.00
	2 2023	Januari	ATM	294	1012552328.00
	3 2023	Januari	Teller	284	969668609.00
	4 2023	Januari	Internet Banking	214	727297330.00
	5 2023	Januari	EDC / Mesin Kasir	78	300451536.00
B	6 2023	Januari	Call Center / IVR	80	281695826.00
	7 2023	Februari	Mobile Banking	448	1451641985.00
	8 2023	Februari	Teller	267	897797861.00
	9 2023	Februari	ATM	261	894938937.00
	10 2023	Februari	Internet Banking	213	779015563.00
	11 2023	Februari	Call Center / IVR	70	267626773.00
	12 2023	Februari	EDC / Mesin Kasir	81	262581098.00
	13 2023	Maret	Mobile Banking	508	1672358427.00
	14 2023	Maret	ATM	314	1095586613.00
	15 2023	Maret	Teller	258	840492298.00
	16 2023	Maret	Internet Banking	173	581727720.00

○ Volume & Nilai Transaksi per Channel per Tahun

#	year	channel_name	volume_transaksi	nilai_transaksi	
P	1	2023	Mobile Banking	5928	19966845883.00
	2	2023	Teller	3376	11246846064.00
	3	2023	ATM	3328	11106080266.00
	4	2023	Internet Banking	2469	8278376140.00
	5	2023	EDC / Mesin Kasir	861	2977506185.00
B	6	2023	Call Center / IVR	815	2741400186.00
	7	2024	Mobile Banking	5925	20053137351.00
	8	2024	Teller	3296	11091994637.00
	9	2024	ATM	3348	11012719407.00
	10	2024	Internet Banking	2499	8240913369.00
	11	2024	Call Center / IVR	842	2747300875.00
	12	2024	EDC / Mesin Kasir	820	2682213208.00
	13	2025	Mobile Banking	5715	19042196966.00
	14	2025	ATM	3365	11107787849.00
	15	2025	Teller	3255	10516168518.00
	16	2025	Internet Banking	2487	8362342797.00

- Trend Volume Transaksi dan Nilai Transaksi antar Channel

#	channel_name	volume_transaksi	nilai_transaksi
1	ATM	10041	33226587542.00
2	Mobile Banking	17568	59062180200.00
3	Internet Banking	7455	24881632306.00
4	Teller	9927	32855009219.00
5	EDC / Mesin Kasir	2538	8534582436.00
6	Call Center / IVR	2471	8221186905.00

- Trend Growth Volume Transaksi Channel perbulan

	#	year	month_name	channel_name	volume_transaksi	prev_volume	volume_growth_pct
P	13	2024	Januari	ATM	282	314	-10.19
	14	2024	Mei	ATM	287	282	1.77
	15	2024	Desember	ATM	272	287	-5.23
	16	2024	Maret	ATM	285	272	4.78
	17	2024	Agustus	ATM	302	285	5.96
B	18	2024	Juni	ATM	267	302	-11.59
	19	2024	Oktober	ATM	282	267	5.62
	20	2024	November	ATM	285	282	1.06
	21	2024	September	ATM	265	285	-7.02
	22	2024	Februari	ATM	249	265	-6.04
	23	2024	April	ATM	287	249	15.26
	24	2024	Juli	ATM	285	287	-0.70
	25	2025	Maret	ATM	288	285	1.05
	26	2025	November	ATM	282	288	-2.08
	27	2025	Juni	ATM	282	282	0.00
	28	2025	Agustus	ATM	270	282	-4.26

2. Customer 360

https://github.com/tryaspane/final_project_data_pipeline_bni_corpu/blob/2790a96ed1a2e8e6fa443562b23f41d69c1fd532/query-training/customer360.sql

- Top Nasabah Paling Aktif (Frekuensi & Nilai)

	#	tomer_id	customer_code	full_name	segment	account_no	account_type	product_name
P	1		CST000792	Yanuar Susanto	RETAIL	1199	TABUNGAN	Tabungan Bisnis
	2	8	CST001698	Yanuar Anggraini	RETAIL	2546	TABUNGAN	Tabungan Bisnis
	3	8	CST001128	Yanuar Lubis	RETAIL	1691	GIRO	Giro Reguler
B	4	8	CST001068	Ivan Prasetyo	RETAIL	1600	TABUNGAN	Tabungan Online
	5	7	CST001717	Ulfa Prasetyo	RETAIL	2574	DEPOSITO	Deposito 6 Bulan
	6	2	CST001742	Aditya Hidayat	RETAIL	2611	DEPOSITO	Deposito 3 Bulan
	7	2	CST001622	Bagas Susanto	RETAIL	2430	TABUNGAN	Tabungan Online
	8		CST000673	Dimas Simanjuntak	PRIORITY	1013	TABUNGAN	Tabungan Reguler
	9	7	CST001327	Zara Santoso	RETAIL	1993	TABUNGAN	Tabungan Bisnis
	10	3	CST001883	Xena Manurung	VIP	2832	TABUNGAN	Tabungan Reguler
	11		CST000456	Qori Gunawan	VIP	685	TABUNGAN	Tabungan Online
10			CST000000	Liana	VIP	000	TABUNGAN	Tabungan

- Distribusi Per Segment

	#	segment	jumlah_nasabah	volume_transaksi	nilai_transaksi	rata_rata_transaksi
P	1	RETAIL	1495	40544	135372068379.00	3338892.77
	2	PRIORITY	242	6901	22898547122.00	3318149.13
	3	VIP	87	2555	8510563107.00	3330944.46

3. Branch Performance

https://github.com/tryaspane/final_project_data_pipeline_bni_corpu/blob/2790a96ed1a2e8e6fa443562b23f41d69c1fd532/query-training/branch_performance.sql

	#	region	branch_id	branch_code	branch_name	city	jumlah_transaksi	total_nilai_transaksi	rank_volume	rank_nilai
P	1	BALI-NUSA	25	KCP-DPS02	KCP Kuta Legian	Badung	1744	5953376477.00	1	1
	2	BALI-NUSA	24	KCU-DPS01	KCU Denpasar Gajah Mada	Denpasar	1609	5229802031.00	2	2
B	3	JABAR	10	KCP-CRM01	KCP Cirebon Kartini	Cirebon	2155	7414651702.00	1	1
	4	JABAR	9	KCP-BDG02	KCP Bandung Dago	Bandung	1921	6487002869.00	2	2
B	5	JABAR	8	KCU-BDG01	KCU Bandung Asia Afrika	Bandung	1862	6130300034.00	3	3
	6	JABODETABEK	2	KCU-JKT02	KCU Jakarta Selatan	Jakarta	2117	6942196679.00	1	2
B	7	JABODETABEK	7	KK-BGR01	KK Bogor Pajajaran	Bogor	2103	7023163801.00	2	1
	8	JABODETABEK	3	KCU-JKT03	KCU Jakarta Utara	Jakarta	1995	6630400109.00	3	3
B	9	JABODETABEK	6	KCP-DEK01	KCP Depok Margonda	Depok	1951	6368789477.00	4	5
	10	JABODETABEK	5	KCP-TGR01	KCP Tangerang Selatan	Tangerang	1939	6475539012.00	5	4
B	11	JABODETABEK	4	KCP-BKS01	KCP Bekasi Jaya	Bekasi	1783	5792004142.00	6	6

4. Channel Analysis

https://github.com/tryaspane/final_project_data_pipeline_bni_corpu/blob/2790a96ed1a2e8e6fa443562b23f41d69c1fd532/query-training/channel_analysis.sql

	#	channel_name	channel_category	volume_transaksi	nilai_transaksi	rata_rata_transaksi
P	1	Mobile Banking	DIGITAL	17568	59062180200.00	3361918.27
	2	ATM	PHYSICAL	10041	33226587542.00	3309091.48
	3	Teller	PHYSICAL	9927	32855009219.00	3309661.45
	4	Internet Banking	DIGITAL	7455	24881632306.00	3337576.43
	5	EDC / Mesin Kasir	PHYSICAL	2538	8534582436.00	3362719.64
B	6	Call Center / IVR	DIGITAL	2471	8221186905.00	3327068.76

5. Product Performance

https://github.com/tryaspane/final_project_data_pipeline_bni_corpu/blob/2790a96ed1a2e8e6fa443562b23f41d69c1fd532/query-training/product_performance.sql

	#	year	month	month_name	product_name	account_type	volume_transaksi	nilai_transaksi	saldo_rata_rata	rata_rata_nominal_transaksi
P	1	2023	1	Januari	Tabungan Junior	TABUNGAN	231	812142779.00	49600532.43	3515769.61
	2	2023	1	Januari	Tabungan Bisnis	TABUNGAN	222	813341553.00	51222176.59	3663700.69
	3	2023	1	Januari	Tabungan OnLine	TABUNGAN	214	765403500.00	49993454.60	3576651.87
	4	2023	1	Januari	Tabungan Reguler	TABUNGAN	187	652827989.00	48858064.18	3491958.76
B	5	2023	1	Januari	Giro Valas	GIRO	109	339042011.00	50848275.52	3110477.17
	6	2023	1	Januari	Deposito 1 Bulan	DEPOSITO	97	316975774.00	52270557.54	3267791.48
	7	2023	1	Januari	Giro Reguler	GIRO	95	344958160.00	51200030.26	3631138.53
	8	2023	1	Januari	Giro Bisnis	GIRO	95	327156729.00	52418679.28	3443755.04
	9	2023	1	Januari	Deposito 12 Bulan	DEPOSITO	66	228551705.00	54215544.85	3462904.62
	10	2023	1	Januari	Deposito 6 Bulan	DEPOSITO	61	189394969.00	45055563.23	3104835.56
	11	2023	1	Januari	Deposito 3 Bulan	DEPOSITO	59	255378694.00	52970178.64	4328452.44
	12	2023	2	Februari	Tabungan OnLine	TABUNGAN	214	762477070.00	51106709.98	3562976.96

6. Fraud Detection

https://github.com/tryaspane/final_project_data_pipeline_bni_corpu/blob/2790a96ed1a2e8e6fa443562b23f41d69c1fd532/query-training/risk_fraud_detection.sql

- Transaksi dengan Nominal Tidak Wajar (transaksi >3x

	#	transaction_id	transaction_code	customer_code	full_name	channel_name	transaction_at	amount	avg_amount	kellip
P	1	34357	TRX0000034357	CST001300	Diana Hartono	Mobile Banking	2024-01-27 02:17:47	9999892.00	3209465.471698113208	3.12
	2	49048	TRX0000049048	CST000092	Feni Siregar	Mobile Banking	2023-08-30 11:57:29	9997435.00	2940210.000000000000	3.40
	3	36295	TRX0000036295	CST001167	Hendra Siregar	Internet Banking	2025-05-15 18:30:10	9995971.00	2742797.411764705882	3.64
	4	25544	TRX0000025544	CST001650	Nita Harahap	Teller	2023-04-14 15:40:33	9994091.00	3299349.382978723404	3.03
B	5	2229	TRX000002229	CST001584	Sandi Prasetyo	Teller	2024-01-19 14:11:26	9902338.00	2907019.534083720930	3.44
	6	27470	TRX0000027470	CST000171	Eka Manurung	ATM	2023-01-01 07:45:52	9992210.00	3208736.826006956522	3.11
	7	37308	TRX0000037308	CST001104	Fajar Permata	EDC / Mesin Kasir	2023-11-03 04:40:10	9991866.00	3145677.875000000000	3.18
	8	17743	TRX0000017743	CST001542	Bagas Nasution	ATM	2024-02-21 10:37:05	9909557.00	3236639.000000000000	3.09
	9	6242	TRX0000006242	CST000082	Bella Panjaitan	ATM	2024-08-30 19:09:08	9988992.00	3058707.297297297297	3.27
	10	8234	TRX0000008234	CST001564	Oori Simanjuntak	Mobile Banking	2025-10-30 13:57:55	9987250.00	2998374.473684210526	3
	11	15753	TRX0000015753	CST001929	Joko Manurung	Mobile Banking	2025-10-24 12:12:51	9983668.00	2705096.666666666667	3

- Transaksi gagal berulang (>=5 kali)

	#	customer_code	full_name	channel_name	jumlah_failed
P	1	CST000666	Siti Gunawan	Mobile Banking	9
	2	CST000188	Oki Nasution	Mobile Banking	8
	3	CST000475	Qita Siregar	Mobile Banking	7
	4	CST001621	Rina Gunawan	Mobile Banking	7
	5	CST001489	Zaki Kusuma	Mobile Banking	7
B	6	CST000682	Fitri Sihombing	Mobile Banking	7
	7	CST000718	Eka Siregar	Mobile Banking	7
	8	CST000879	Maman Harahap	Mobile Banking	7
	9	CST001433	Sandi Siregar	Mobile Banking	6
	10	CST000812	Olga Lubis	Mobile Banking	6
	11	CST000748	Ayu Susanto	Mobile Banking	6
	12	CST001418	Zaki Panjaitan	Mobile Banking	6
	13	CST000393	Doni Mulyono	Mobile Banking	6
	14	CST000667	Sandi Santoso	Mobile Banking	6
	15	CST001778	Sari Hermawan	Mobile Banking	6
	16	CST000695	Rudi Nugroho	Mobile Banking	6